LIVESTOCK AI MANAGER (LAIM)

A Data-Driven Decision Support System for Precision Artificial Insemination in Kenya

1. INTRODUCTION & BACKGROUND

Agriculture is the backbone of the Kenyan economy, contributing approximately 33% of the GDP. The dairy sector, specifically, is a critical source of livelihood for millions of smallholder farmers in regions such as Kakamega, Kiambu, and Uasin Gishu. However, the sector faces significant challenges in breeding efficiency. Many farmers rely on intuition rather than data when scheduling Artificial Insemination (AI), leading to low conception rates and high veterinary costs.

2. PROBLEM STATEMENT

The current success rate of Artificial Insemination in smallholder farms often falls below 40%. Key contributing factors include:

1. Inaccurate detection of the Estrus cycle (Heat).
2. Poor matching of Sire (Bull) genetics to the specific Cow breed.
3. Lack of historical record-keeping to track reproductive health.

Failed inseminations result in wasted semen straws (approx. KES 1,500 - 4,000 per attempt), loss of milk production time, and extended calving intervals.

3. THE INNOVATION: LAIM

Livestock AI Manager (LAIM) is a web-based platform that leverages Machine Learning (XGBoost) to analyze biological metrics and predict the probability of conception success before the farmer pays for the service.

Key Technical Features:

- Predictive Engine: Analyzes Age, Weight, BCS, and Activity Index to calculate a success percentage.
- Genetic Catalog: A digital database of Kenyan bulls (Friesian, Sahiwal, Boran) with fertility scores.
- Digital Records: Automated history logs for auditing past insemination attempts.
- Support System: Integrated ticketing system for farmer technical support.

4. PROJECT OBJECTIVES

- To increase AI conception rates in pilot farms by at least 25%.
- To reduce the financial burden on farmers by preventing untimely inseminations.
- To digitize breeding records for better herd management and genetic tracking.
- To promote the use of data-driven agriculture in rural Kenya.

5. TARGET AUDIENCE

- Small to Medium-scale Dairy Farmers (1-50 cows).
- Veterinary Officers and AI Technicians.
- Agricultural Cooperatives and Dairy Societies.

6. CONCLUSION

LAIM represents a significant step towards modernizing dairy farming in Kenya. By bridging the gap between biological data and actionable insights, the project aligns with the national goals of food security and economic empowerment for the agricultural workforce.